Trainings

Note: A video supports the information provided in this procedure. Read the procedure prior to viewing the video.

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919588
- Vehicle Plumbing Clean Care Kit, Part Number 4919425
- EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437
- Air Pressure Regulator Kit, Part Number 3164231, or equivalent.

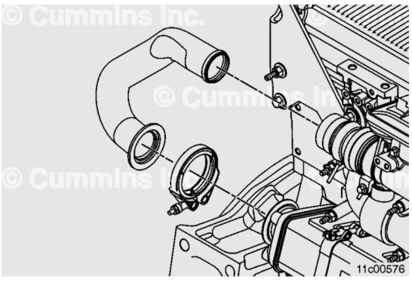
Additional Service Items

- Fleetguard® 3-Way Coolant Test Kit, CC2602B

Coolant Strip Test

⚠ WARNING ⚠

The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair. Make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components.



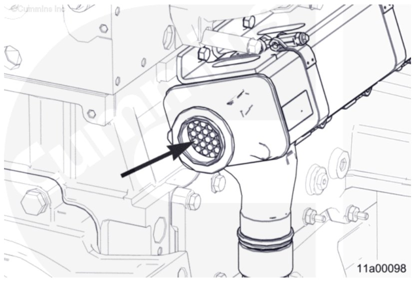
Perform the exhaust gas recirculation (EGR) cooler coolant strip test with the EGR cooler on engine. Use Fleetguard® 3-Way Coolant Test Kit, CC2602B. Follow the instructions on the coolant test strip package.

Note : It may be necessary to test raw coolant from the cooling system to be sure the Fleetguard® 3-Way Coolant Test strips will react if the type of coolant is unknown.

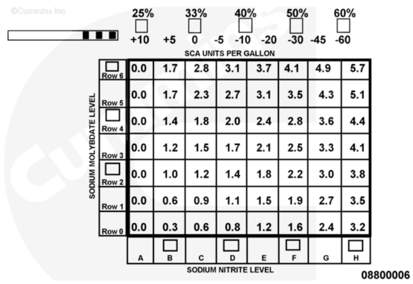
Coolant Test Strip - EGR Cooler Exhaust Outlet

- Test the Fleetguard® 3-Way Coolant Test Strip in a small, clean container with 25 ml (0.8 oz) of distilled water to create a baseline for analysis. Save the test strip for comparison.
- Remove rear EGR connection tube. Refer to Procedure 011-069 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-069.html)

- Remove a small sample of soot from EGR cooler exhaust outlet with a clean tool.
- Place the soot sample into the existing small container containing the 25ml (0.8 oz) of distilled water. Mix the soot into the distilled water.

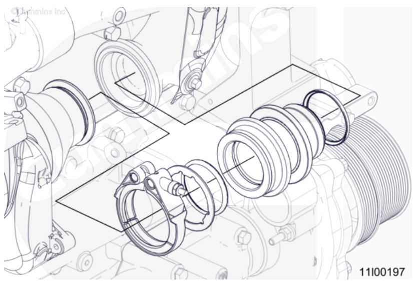


- Test the soot and water solution with a Fleetguard® 3 Way Coolant Test strip.
- The EGR cooler is reusable if **all** pads on test strip show the same color as the distilled water **only** test strip.
- Install the exhaust outlet tube and return to published troubleshooting.
- A change in any of the three test pads indicates coolant is in the exhaust outlet side of the EGR cooler. Proceed to inlet side test.



	A	B	C	D	E	F	G	H
Row 6	0.0	1.7	2.8	3.1	3.7	4.1	4.9	5.7
Row 5	0.0	1.7	2.3	2.7	3.1	3.5	4.3	5.1
Row 4	0.0	1.4	1.8	2.0	2.4	2.8	3.6	4.4
Row 3	0.0	1.2	1.5	1.7	2.1	2.5	3.3	4.1
Row 2	0.0	1.0	1.2	1.4	1.8	2.2	3.0	3.8
Row 1	0.0	0.6	0.9	1.1	1.5	1.9	2.7	3.5
Row 0	0.0	0.3	0.6	0.8	1.2	1.6	2.4	3.2

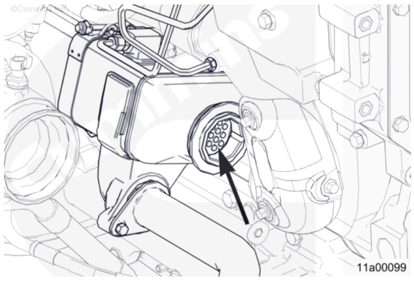
Coolant Test Strip - EGR Cooler Exhaust Inlet



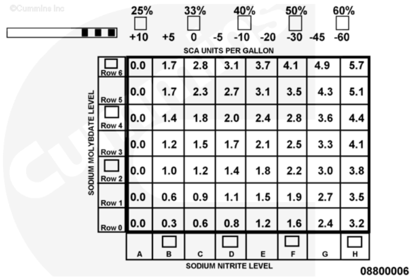
- Make sure to use a new small container or wash the existing small container with soap and water to remove any residue from previous test.
- Test the Fleetguard® 3-Way Coolant Test Strip in a small clean container with 25 ml (0.8 oz) of distilled water to create a baseline for analysis. Save the test strip for comparison

- Remove front EGR cooler connection. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)
- Make sure to clean the tool from the previous test.

- Remove small sample of soot from EGR cooler exhaust inlet with a clean tool.
- Place soot sample into the existing small container containing the 25ml (0.8 oz) of distilled water. Mix the soot into the distilled water.



- Test soot and water solution with a Fleetguard® 3-Way Coolant Test strip.
- A change in any of the three test pads which is different from the distilled water **only** test strip indicates coolant is in the exhaust inlet side of the cooler. Proceed to Preparatory, Remove, and Pressure Test Steps.
- The EGR cooler is malfunctioned if coolant is in the outlet side and **not** in the inlet side of the EGR cooler.



Coolant Test Strip Results Guide		
Coolant in outlet	**Coolant in inlet**	**Outcome**
No	N/A	EGR cooler is **not** malfunctioning. Return to published troubleshooting.
Yes	No	EGR cooler is malfunctioning. Replace EGR cooler.
Yes	Yes	Pressure test the EGR cooler.

Preparatory Steps

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

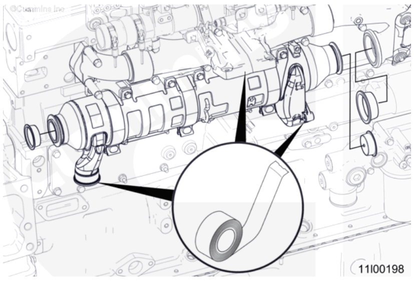
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Brush away all loose dirt from around the area of the air handling connections to reduce the possibility of contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Remove the air intake pipe to the turbocharger. See equipment manufacturer service information. Use protective caps from the Vehicle Plumbing Clean Care Kit, Part Number 4919425, to cover the connection points.
- Remove the aftertreatment adapter pipe. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-043.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-031.html)
- Remove the EGR cooler connection. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)
- Remove the rear EGR connection tube. Refer to Procedure 011-069 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-069.html)

Remove

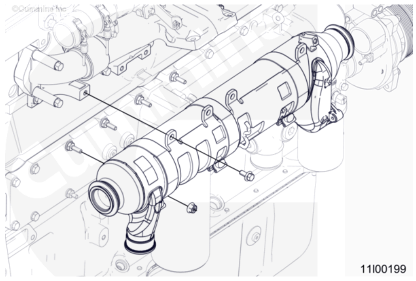
Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, to cover the open ports on the EGR cooler and exhaust manifold.



Remove the capscrews that attach the EGR cooler support bracket to the engine block and the EGR cooler mounting nuts on the lubricating oil cooler assembly.

Remove the EGR cooler assembly from the engine.

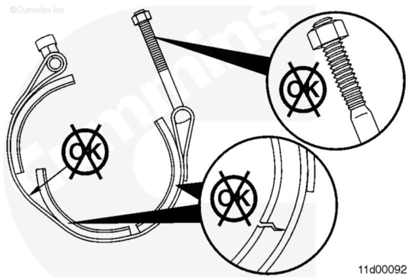
Remove EGR cooler support bracket.



Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler.



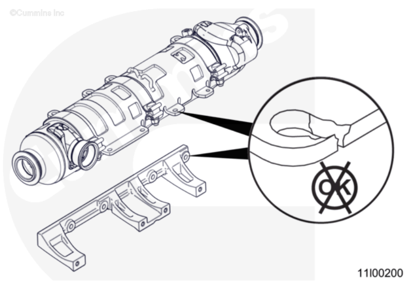
Inspect the V-band clamp threads and EGR cooler connection for damage. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)

Replace the V-band clamp or EGR cooler connection if cracks or other damage is found.

Inspect the EGR cooler and EGR cooler support bracket for cracks or other damage.

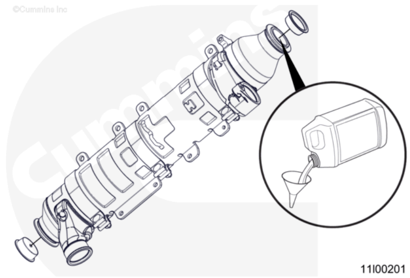
Replace the component if cracks or other damage is found.

Note : For more information regarding inspection of the EGR cooler, refer to EGR Cooler Reuse Guidelines for ISX CM871, ISX12/ISX11.9 CM2250, ISX15 CM2250, QSX15 CM2250 ECF, QSX11.9 CM2250 ECF, PowerGen QSX15 CM2250 ECF, and PowerGen QSX15 CM2250, Bulletin 2883594 (/qs3/pubsys2/xml/en/bulletin/2883594.html).



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Cap the exhaust gas outlet end of the EGR cooler with the plastic caps supplied in the EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437.

Completely fill the EGR cooler with safety solvent.

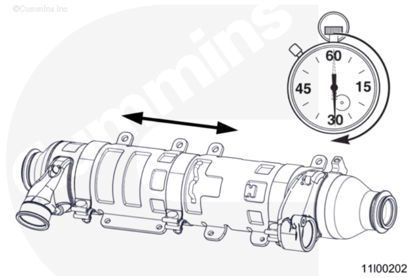
Cap the exhaust gas inlet of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437.

Lay the EGR cooler on its side.

Allow the EGR cooler to soak for approximately 20 minutes.

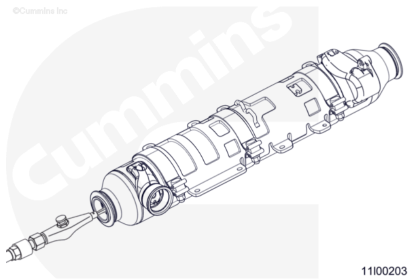
Drain approximately 25 percent of the liquid from the cooler and recap. Dispose of this liquid properly.

Shake the cooler by hand, from end-to-end, for approximately 30 seconds.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Drain the remaining safety solvent from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.

Cap the exhaust gas outlet of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437.

Completely fill the EGR cooler with water.

Cap the exhaust gas inlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437.

Shake the cooler by hand, from end-to-end, for approximately 30 seconds.

Drain the water from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler.

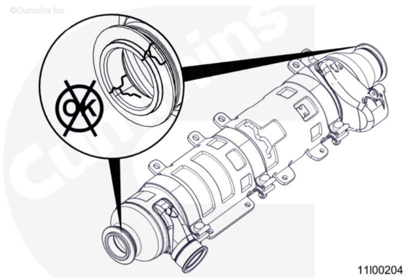
Replace the EGR cooler if cracks or other damage is found.

Clean the EGR cooler exhaust gas inlet and outlet with safety solvent.

Dry with compressed air.

Inspect the EGR cooler exhaust gas inlet and outlet for signs of fretting, cracks, or other damage.

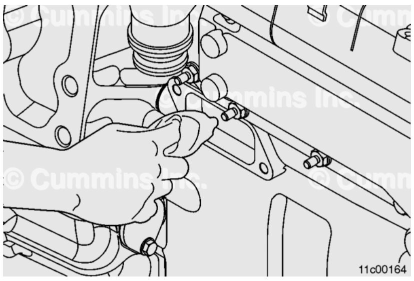
Replace the EGR cooler if any damage is found.



Clean the mounting surfaces with safety solvent to remove any coolant deposits or debris.

Clean the opening in the block where the EGR cooler coolant return elbow connection is installed to remove any coolant deposits or debris. Otherwise, the press-in-place seal joint may leak.

Clean the EGR cooler coolant supply and return couplings to remove any coolant deposits or debris.

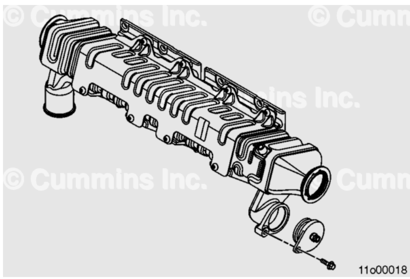


Pressure Test

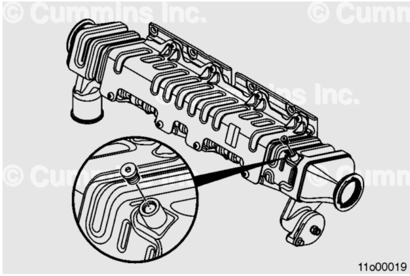
Note : This test is performed with the EGR cooler removed from the engine.

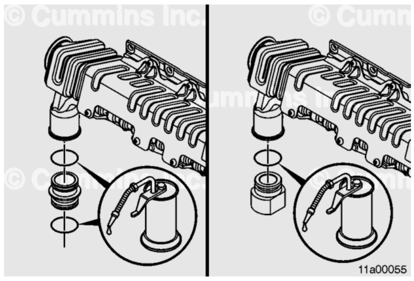
Install the plug, Part Number 4918689, with the o-ring, Part Number 3683814, supplied in the EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437, into the coolant supply port of the EGR cooler.

Secure the plug with a M8 x 1.25 capscrew.



Install the straight thread plug (1), Part Number 4002056, and o-ring, Part Number 3348319, from EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437, into the EGR cooler coolant vent port.





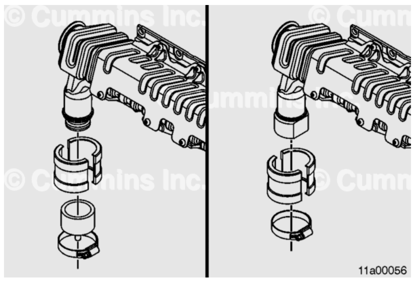
Locate the EGR cooler coolant return coupling, Part Number 3683586.

Inspect the o-rings, Part Number 3683814, for cracks, cuts, or other damage. Replace if necessary.

Lubricate the o-rings with clean engine coolant, soapy water, or vegetable oil to aid in installation.

Install the EGR cooler coolant return coupling into the EGR cooler coolant return port.

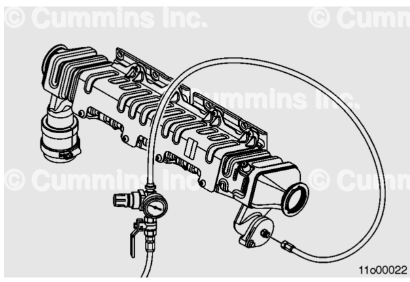
Note : If leak test plug, Part Number 2892396, is available, install it into the EGR cooler instead of the EGR cooler coolant return coupling, Part Number 3683586.



Install the collar, Part Number 4918446, and cap, Part Number 4918444, over the EGR cooler coolant return coupling and EGR cooler coolant return port.

Note : If leak test plug, Part Number 2892396, is available, cap, Part Number 4918444, will not be needed for this step.

Secure with the hose clamp, Part Number 3026396, from EGR Cooler Leak Check Kit, Part Number 4918655 or 2892437.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

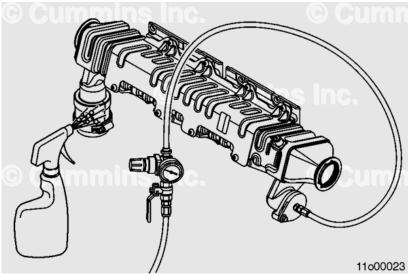
Note : The test **must** be performed when the EGR cooler has reached ambient temperature. Do **not** apply air pressure to a hot EGR cooler.

Note : Air Pressure Regulator Kit, Part Number 3164231, or equivalent, is suitable for use during this step.

Install air pressure regulator, Part Number 3164231, or equivalent, onto the plug, Part Number 4918689, in the EGR cooler coolant supply port.

Install a ball-type shutoff valve onto the supply side of the air pressure regulator to allow the air supply to be shut off during testing.

Make sure the air pressure regulator and ball-type shutoff valve are closed.



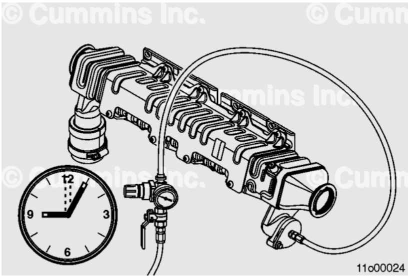
Connect a compressed air supply to the pressure regulator.

Apply 620 kPa [90 psi] of air pressure to the EGR cooler.

Inspect for air escaping from the EGR cooler or pressure test assembly.

Check for leaks by spraying the exterior of the EGR cooler and all lines and fittings with a mixture of mild soap and water. Repair leaking fittings or lines.

Use the ball valve to shut off the air supply to the air pressure regulator, Part Number 3164231, or equivalent.

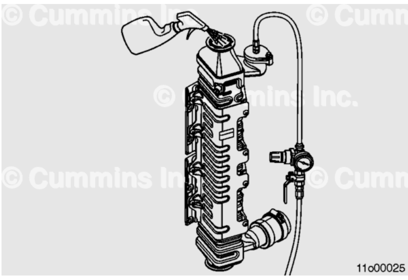


Disconnect the compressed air supply from the air pressure regulator.

Record the time the compressed air supply was shut off and removed from the air pressure regulator.

Record the pressure shown on the air pressure regulator gauge.

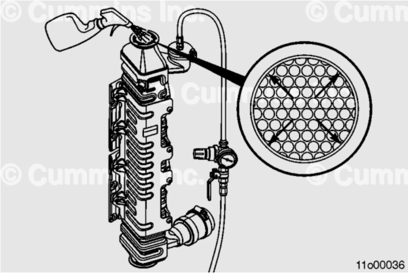
Place the EGR cooler in a vertical position with the exhaust gas inlet pointing upward.



Spray the exhaust inlet bulkhead and gas tubes with a mixture of mild soap and water.

Make sure the inside corners of the bulkhead have been sprayed.

Note : It is important to wet the entire bulkhead with the mixture of mild soap and water. The highest percentage of leaks occur in the inside edge of the EGR cooler.

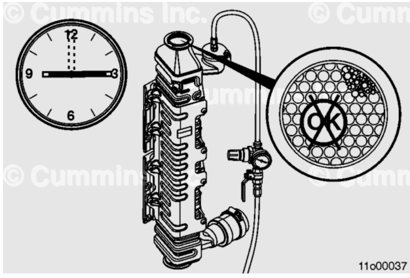


Allow the EGR cooler to remain pressurized for 15 minutes.

Inspect the inside corners of the bulkhead and gas tubes for the continuous formation of bubbles.

Measure and record the air pressure.

If the measured pressure loss in the EGR cooler is more than 103 kPa [15 psi] in 15 minutes, **or** if there is a continuous formation of bubbles near the gas tubes and bulkhead, the EGR cooler is **not** usable.



Note : For more information regarding inspection of the EGR cooler, refer to the EGR Cooler Reuse Guidelines for ISX CM871, ISX12/ISX11.9 CM2250, ISX15 CM2250, QSX15 CM2250 ECF, QSX11.9 CM2250 ECF, PowerGen QSX15 CM2250 ECF, and PowerGen QSX15 CM2250, Bulletin 2883594. (/qs3/pubsys2/xml/en/bulletin/2883594.html)

Note : The pressure test is meant to identify large leaks while the inspection with mild soap and water is meant to identify small leaks. Do not use small losses observed during the pressure test only to evaluate the EGR cooler for reuse, as they are often associated with issues with the leak test equipment.

Note : Stationary bubbles that form on the soot deposits or in the gas tubes are not an indication of a leak.

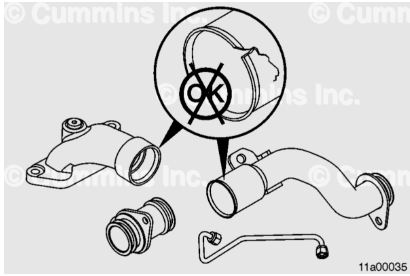
Note : The continuous formation of bubbles is an indication that the EGR cooler is not reusable.

Note : If the EGR cooler is not usable based on a loss of pressure, inspect the air pressure regulator, air lines, ball valves, o-rings, and other components carefully for leaks. A mixture of mild soap and water may be sprayed on the assembly to aid in leak detection.

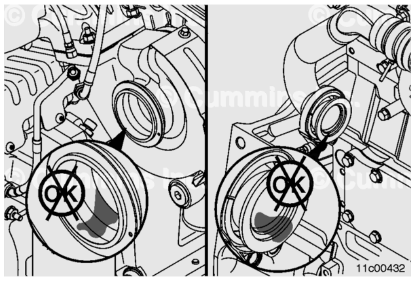
Note : If the EGR cooler is not usable based on a loss of pressure, it may be necessary to evaluate the EGR cooler pressure testing assembly for leaks before replacing the EGR cooler. Assorted fittings or a known good EGR cooler can be used for this test.

If the EGR cooler is usable, inspect the EGR cooler coolant lines, couplings, o-rings, and grommets for cracks or other damage.

Replace the necessary components if cracks or other damage is found. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-031.html)



If the EGR cooler is reusable, remove the components installed from the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.



If the EGR cooler is **not** reusable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits.

Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing.

If coolant deposits are found in the turbocharger turbine housing, inspect the turbocharger, EGR valve, EGR mass measurement flow assembly, exhaust transfer connection elbow, EGR crossover tube, and the crankcase pressure sensor for progressive damage.

Reference the following procedures:

- Turbocharger: Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- EGR valve: Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-022.html)
- EGR mass measurement flow assembly: Refer to Procedure 011-066 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-066.html)
- Exhaust transfer connection tube: Refer to Procedure 011-068 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-068.html)
- EGR crossover tube: Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html)
- Crankcase pressure sensor: Refer to Procedure 019-445 in Section 19. (/qs3/pubsys2/xml/en/procedures/377/377-019-445.html)

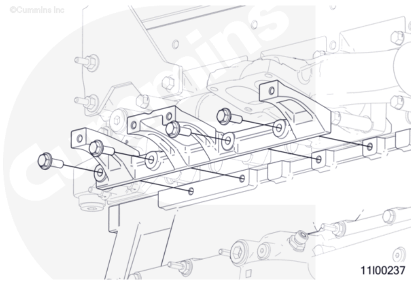
Note : **If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, it could be necessary to clean the charge air cooler. See equipment manufacturer service information.**

Note : **If damage resulted in a hydraulic lock, it could be necessary to remove the injectors and evacuate coolant from the cylinder(s). Refer to Procedure 006-026 in Section 6.**

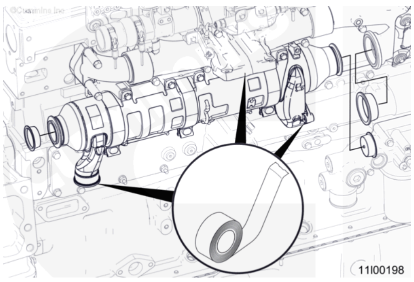
Install

Install the EGR cooler support bracket to the engine block with the capscrews.

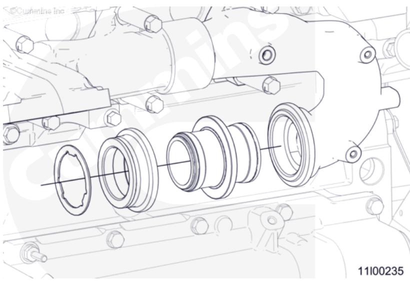
Torque Value: 47 n•m [35 ft-lb]



Remove the protective caps from the Air Handling Clean Care Kit, Part Number 4919588, used to cover the open ports on the EGR cooler and exhaust manifold.



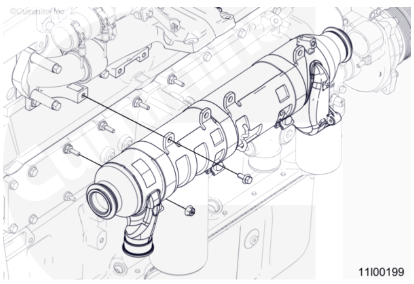
Install the EGR connection tube.



Apply a film of high temperature anti-seize compound to the inside of the V-band clamp to make sure of proper loading on the clamp.

Install a V-band clamp onto the EGR cooler exhaust collar.

Install a new gasket on the pilot end of the EGR cooler where it connects to the exhaust collar.



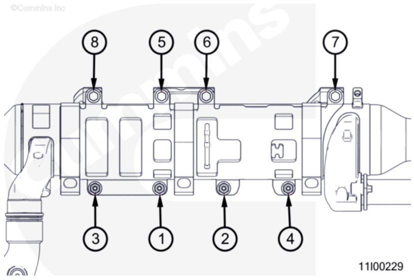
Install the EGR cooler on the studs in the lubricating oil cooler assembly.

Install the EGR cooler mounting capscrews at the top of the EGR cooler mounting bracket.

Install and handtighten the mounting nuts.

Install and handtighten the mounting nuts.

Tighten the capscrews and nuts in the sequence shown.

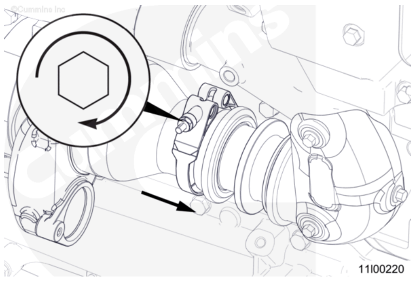


Torque Value:
Capscrews 47 n•m [35 ft-lb]

Torque Value:
Nuts 33 n•m [24 ft-lb]

⚠ CAUTION ⚠

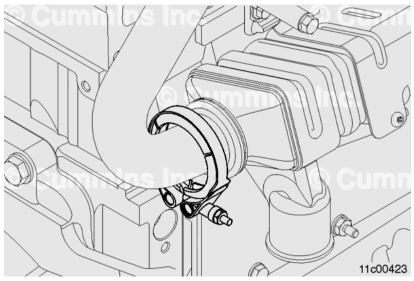
Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be able to be reused.



Position the V-band clamp onto the EGR cooler exhaust gas inlet connection.

Torque Value: 17 n•m [150 in-lb]

Install a new gasket on the pilot end of the EGR cooler exhaust gas outlet where it connects to the rear EGR connection tube.



Position the V-band clamp at the rear EGR connection tube and EGR cooler exhaust gas outlet.

Torque Value: 17 n•m [150 in-lb]

Finishing Steps

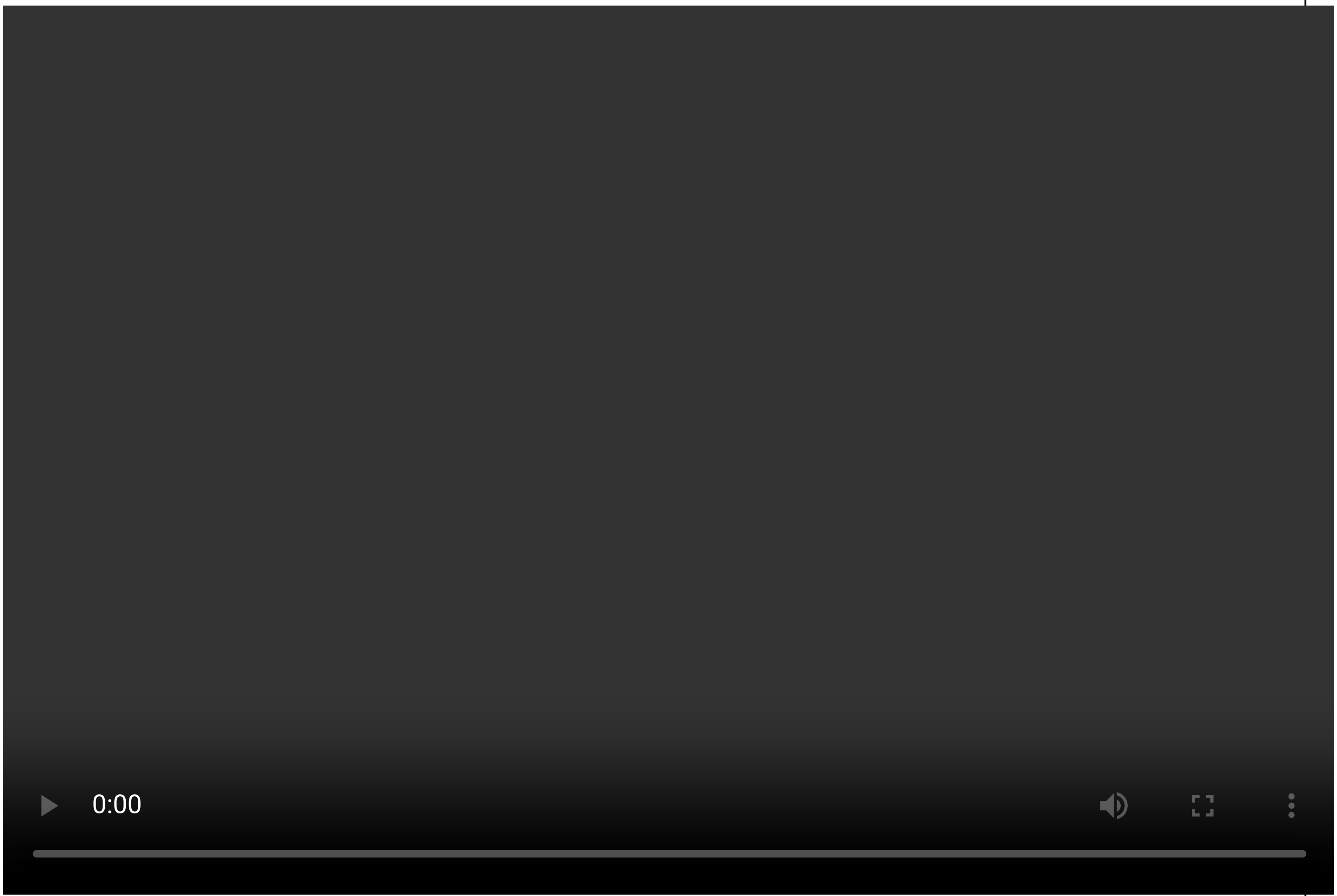
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-031.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Install the aftertreatment adapter pipe. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-043.html)
- Install the air intake pipe to the turbocharger. See equipment manufacturer service information.
- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and verify that all fault codes are inactive.

Note : If a malfunction resulted in coolant entering the exhaust system, inspect the diesel oxidation catalyst inlet for face plugging. Reference the Service Bulletin, Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 (/qs3/pubsys2/xml/en/bulletin/4021600.html). If found to be face plugged, clean the diesel oxidation catalyst with compressed air. Refer to Procedure 011-049 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-049.html)

Note : If a malfunction resulted in coolant entering the exhaust system, perform the Aftertreatment Diesel Particulate Filter Restriction Test to determine if the diesel particulate filter (DPF) needs to be replaced. Refer to Procedure 011-085 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-085.html)



Last Modified: 27-Aug-2018